

Group/Team Name: Life Is Not That Deep

Team member name	Student ID	individual overall work contribution (%)	Signature
Student1: Mohammed Sarim Shaikh	001499385-6	16.66%	MSS
Student2: Gabriel Gheorghiu	001413253-2	16.66%	GG
Student3: Aslim Gulvardar	001490395-4	16.66%	AG
Student4: Martin Glazer	001456453-x	16.66%	MG
Student5: Tej Gohil	001501430-4	16.66%	TG
Student6: Bhumi Poddar	001447331-3	16.66%	BP
Total100%			

Task 1:

Legal, Social, Ethical and Professional Issues (LSEPI)

When building SafeCare's new hospital system, there may be a few serious things to think about like how it affects people apart from the technology side of it. Since it deals with private medical data and access to restricted areas, we must make sure it's handled responsibly from every angle.

Legal:

A very big legal concern is data protection. The system will be handling sensitive details about patients, visitors, and staff, so it has to like fully to comply with GDPR and the UK's Data Protection laws. People using the app should be told exactly what data is being collected, why the being data collected is needed and who all can access it. Since the hospital plans to use things like biometric logins and encryptions, securely storing this data is crucial. Any data breach wouldn't just break the law but would also damage the hospital's reputation and reduce trust as well.

Social:

On the social side, not everyone is tech-savvy. Especially elderly people. They might find the app confusing. So, SafeCare should offer help desks or keep a manual check-in option. Another problem is that the staff is adjusting to the new system. Many are used to paper forms and talking to people in person rather than using online systems, so proper training is crucial too. The tech should make their jobs easier and not take away the human touch that patients value.

Ethical:

When it comes to ethics, privacy and respect for patients must come first. CCTV cameras and sensors need to be installed everywhere, but too many can make people feel like they are being constantly watched. The hospital should only collect the data that is necessary and be clear about why it is being collected. Using patients' personal data for anything other than what they agreed to is wrong; for instance, using it for staff reviews or marketing. Additionally, the system should be

easy for everyone to use, including people with disabilities or not being able to speak English.

Professional:

Regarding professional concerns, those who are designing and running the system must follow proper standards like the BCS Code of Conduct. This means being honest and repeatedly testing everything in the system. Even a small mistake in a hospital system can lead to serious issues for everyone. Good documentation and teamwork are essential to keep things transparent and reliable.

Task 2:

Problem

- The hospital is suffering from poor coordination because the system is isolated and heavily manual.
- Illegal access to off-limits areas like ICUs and operating theatres is a security risk.
- Manual tracking of patients and visitors leads to inaccuracies and delays.
- Quality visitor management is weak, and it is not easy to validate visitor identity and purpose.
- Virtually fractured data systems hamper decision-making and emergency response.
- City growth increases rapidly so as a result there will be high patient volume.

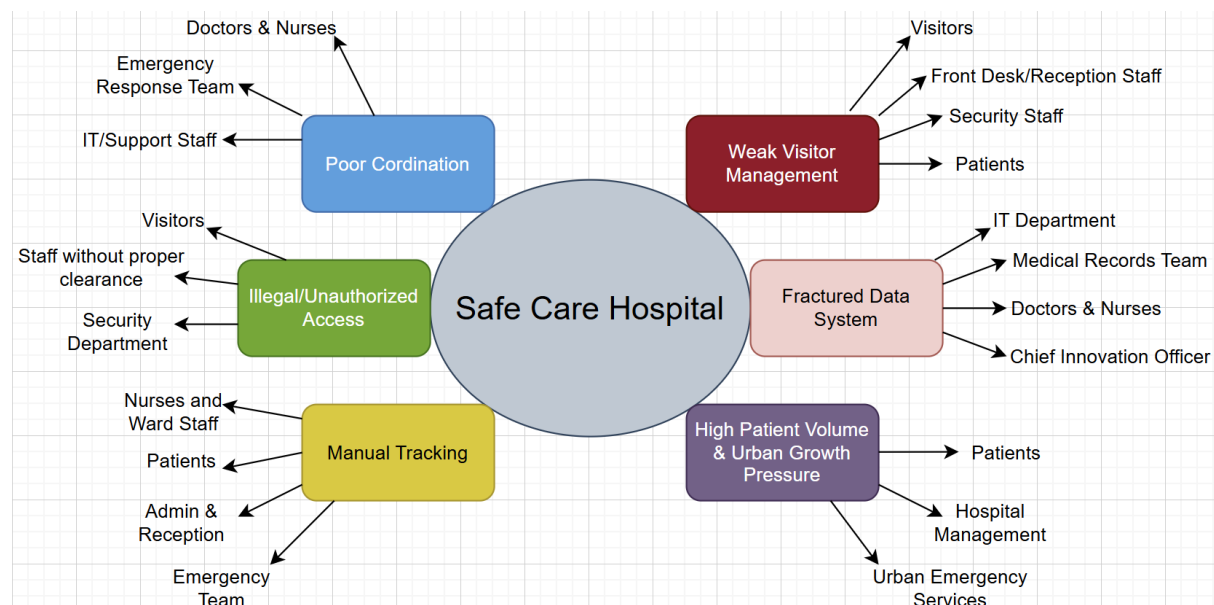


Figure 1 shows the rich picture of the problems

Process

To identify the best approach to create the system, two models of processes are compared:

1. Waterfall Model:

Follows a linear process requirement which includes design, implementation, testing, and deployment.

The Waterfall model is an old, traditional way of developing software, where you must do one step at a time in order. This method is best applied when the project is clear and not likely to change much. It is easy to apply and follow because each stage of the project must be done before the next one starts, so the project is clear and visible how the work is moving forward. (Adenowo and Adenowo)

- Advantages of using the waterfall model are a clearly defined structure, easy to track progress, and suitable when requirements are not fluctuating.
- Some disadvantages: if needs change, the waterfall model is rigid, has delayed feedback, and is slow to adapt to user needs.

2. Agile Model (Scrum):

The app system, which will be created, needs to be done in small parts, getting the customers' feedback frequently.

The Agile model is a newer, better way of developing software that requires flexibility to accommodate change and that works as a team. Rather than the project being finished all in one, the work is split into parts called sprints. This way, the Agile model allows developers to talk more often with the customer and get feedback, enabling them to make quick changes and keep improving the software as they go. This way, it helps the team build better results and is easy to adjust if something new needs to be added to the system. (Wiley and Ambler)

- Some advantages of the model will be that it encourages collaboration, is flexible and can be tested with the user after each phase that has been worked on.
- Some of the disadvantages are that it requires high communication level skills, more time spent in meetings, and it often falls behind schedule.

• Chosen Process: Agile (Scrum)

- Best for changing the needs of the hospital, which allows doctors and administrators to provide feedback during the development of the project.
- We can implement the system in phases, for example, the first access control module, followed by the patient tracking, then emergency alerts and so on.
- Project roles:
 - **The owner** of the application, Chief Innovation Officer, Mr Rahul Singh, sets the main goals and priorities.
 - **The Project Manager** will set tasks based on Agile model ideas and timelines given by the client.

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- **Developers** will create a mobile app that monitors the system and how it connects with other services.
- **The UX Designer** will have to create a simple, user-friendly interface for patients, visitors, and staff so everyone can use the app easily.
- **QA Tester** is testing features for accuracy, security, and fault tolerance.
- **End Users** will be hospital staff, patients, and visitors, who will offer regular feedback.

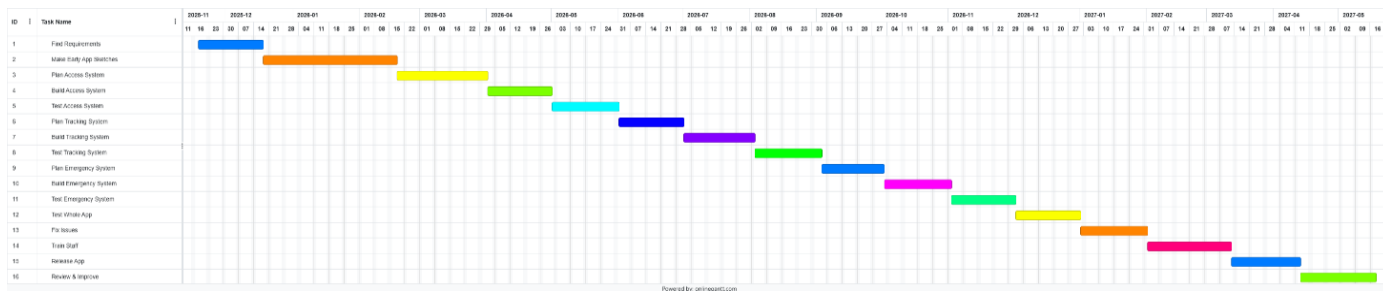
- **Software process on Agile model:**

1. Planning: Identify the problem, why we need this new app and what problems the hospital wants to solve.
2. Analysis: Identify data flow, keep all data safe and private and establish who is allowed to use the data.
3. Design: Make design prototypes of the mobile app and the control panel.
4. Implementation: Work on the project in short steps, test after each version release.
5. Testing: Test the access control, the data privacy, and the emergency alerts.
6. Deployment: Train staff how to use the system, release the app, monitor the usage to make sure it is working well and meet client needs.
7. Maintenance: Fix bugs, improve features according to feedback.

Project

- Goal: Build a secure, smart hospital management system that improves patient safety, visitor control and emergency coordination.
- Budget: £1,800,000 approved by the hospital board.

Total £1,800,000 Budget			
Project team	Number of people	Budget	Notes
Project Manager	1	£180,000	Oversees planning, coordination, and reporting
Developers	5	£750,000	Full-stack team handling backend, frontend, and integration
UX Designer	1	£120,000	Designs interface and ensure good user experience
QA Tester	1	£100,000	Conducts testing and quality assurance
Hardware	-----	£650,000	Includes servers, cameras, and security systems.



- 18-Month Timeline for SafeCare Hospital System Project

Figure 2 shows a Gannt Chart of 18 months timeline

- Resources:
 - Skilled development team (engineers, designers, testers).
 - Hospital IT infrastructure for integration.
 - Provide training so staff and administrators know how to use the system.
 - Safe servers to store patient and visitor information.
- Milestones:
 - Months 1 to 3: Requirements and prototype.
 - Months 4 to 6: Access control and registration system.
 - Months 7 to 9: Visitor and patient tracking module.
 - Months 10 to 12: Emergency alert and monitoring system.
 - Months 13 to 15: Testing, bug fixing, and staff training.
 - Months 16 to 18: Final deployment and evaluation.

Product

- Deliverables (artefacts):**
 - Mobile app for patients and visitors with digital badge/QR code access.
 - Centralised hospital monitoring dashboard with real-time tracking.
 - Database system for storing and managing patient, staff, and visitor data.
 - RFID and biometric access control features.
 - Emergency alert and evacuation automation.
 - Source code, system documentation, and training manuals.

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- The system is designed to provide hospital managers with reports and data, including records of logins and activity, as well as safety information.
- User Manual
- **Quality goals:**
 - System must be secure (GDPR compliant, encrypted data).
 - Always works and available 24/7.
 - User friendly interface for all age groups and digital abilities.
 - Scalable for future expansion to multiple hospitals.

People

- **Key stakeholders:**
 - *Chief Innovation Officer (Rahul Singh)* is the project sponsor and decision maker.
 - *Hospital Operations Director (Fiona Clarke)* – oversees day-to-day impact and staff adaptation.
 - *IT/Development Team* software engineers, UX designers, QA testers.
 - *Medical Staff and Nurses, who are end users*, provide feedback on the usability of the system.
 - *Patients and Visitors are* app users who interact with the system.
 - *Security Staff who are* responsible for physical access control and monitoring.
 - *External Vendors* are the hardware suppliers for RFID, cameras, and sensors.
- **Team building the software:**
 - Product Owner, Scrum Master, 3 to 4 software engineers, one UX designer, one QA tester, and one data analyst.
 - Collaboration between technical and hospital staff is essential to ensure practicality and user satisfaction.

Task 3:

Functional Requirements:

Pre-Registration: The system should allow entry of the patient's identity, visit purpose, and contact information, along with their relatives' information, via the mobile application.

QR Code System: The system should automatically create a unique QR code for each person, after their successful registration, which includes where the patient should go within the hospital and the visitor's access permission (which department they can access) enter, for how long they can stay, visiting hours) and it can be scanned with readers at the doors or with the hospital system, and display it in the user's mobile application.

Identity Verification: To preserve order within the hospital, the system should require all staff, visitors, and patients to use RFID cards and QR codes for access to secure areas.

Visitor Time Limit: If a visitor has overstayed, the system should send a warning via the visitor's mobile application and inform security.

Real-Time Location Tracking: The system should display the real-time location (room/department) of patients and visitors to authorised personnel. Patients and their relatives should also be able to see the location of authorised personnel on a central panel.

Emergency Response: The system should be able to automatically share the exact spot of the patient, and the most suitable response team when it receives a signal from a patient's wearable sensor for a significant event, such as a fall or abnormal heart rate, or from an emergency button triggered by staff, so that the patient can be treated as soon as possible. **Emergency Procedure System:** In the event of a critical incident, the system must immediately inform authorities, patients, and their relatives; automatically initiate the safest evacuation protocols; and guide staff and patients along the safest routes.

Instant Alert System: The system must send notifications via the application to authorities for emergency incidents such as unauthorised access attempts, exceeded visit durations, or potential attacks, ensuring quick response.

Non-Functional Requirements:

Performance: The system should be able to support a high volume of patients and operate at a capacity of 700 beds simultaneously.

Security: All sensitive medical data, such as location data or records, must be kept safe and securely stored in accordance with industry-standard protocols.

Privacy/Compliance: The system needs to completely conform to the General Data Protection Regulation and relevant data protection regulations to ensure the confidentiality of patient data. 4. **Time Constraint:** The project in its entirety, with the implementation of the new system, must be completed within the exact 18-month period given by Mr Singh.

Trainability: Before deployment, it should be easy to fully train all staff so that the system will be adopted and used effectively.

Entity Relationship Diagram(ERD):

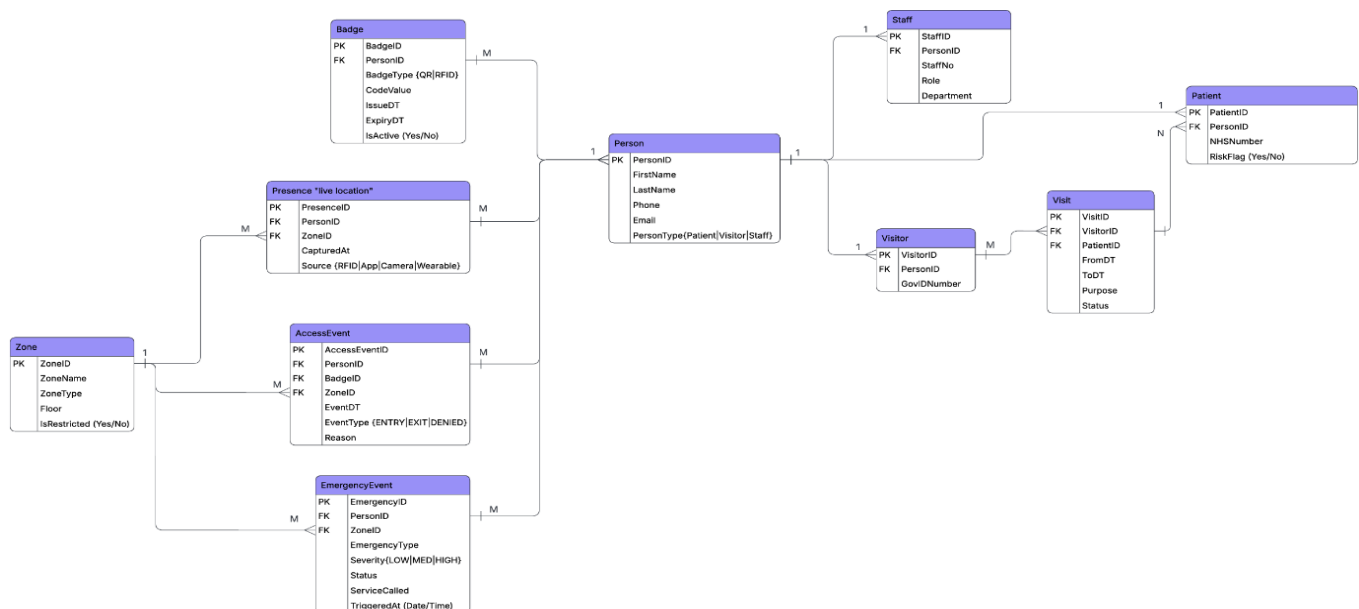


Figure 3 show the Entity Relationship Diagram

The ERD describes the data model of the SafeCare Hospital Monitoring System. The principal actors, include patients, visitors, and staff connected through a unified base entity, which consolidates shared attributes such as name and contact information. Role specific entities such as Patient, Visitor, Staff capture one-to-one relationships, government identification, or staff-specific data. Zones represent different areas of the hospital and distinguish clinical and public areas, floors. Patient and Visitor location updates from RFID, mobile applications, and wearable devices. Emergency Event records critical incidents, including severity, service calls, and timestamps, thereby facilitating immediate response workflows. The Visit table simply links visitors to patients. Overall, the diagram outlines how the system helps the SafeCare Hospital to monitor patients, guests and their activities.

Data Flow Diagram(DFD):

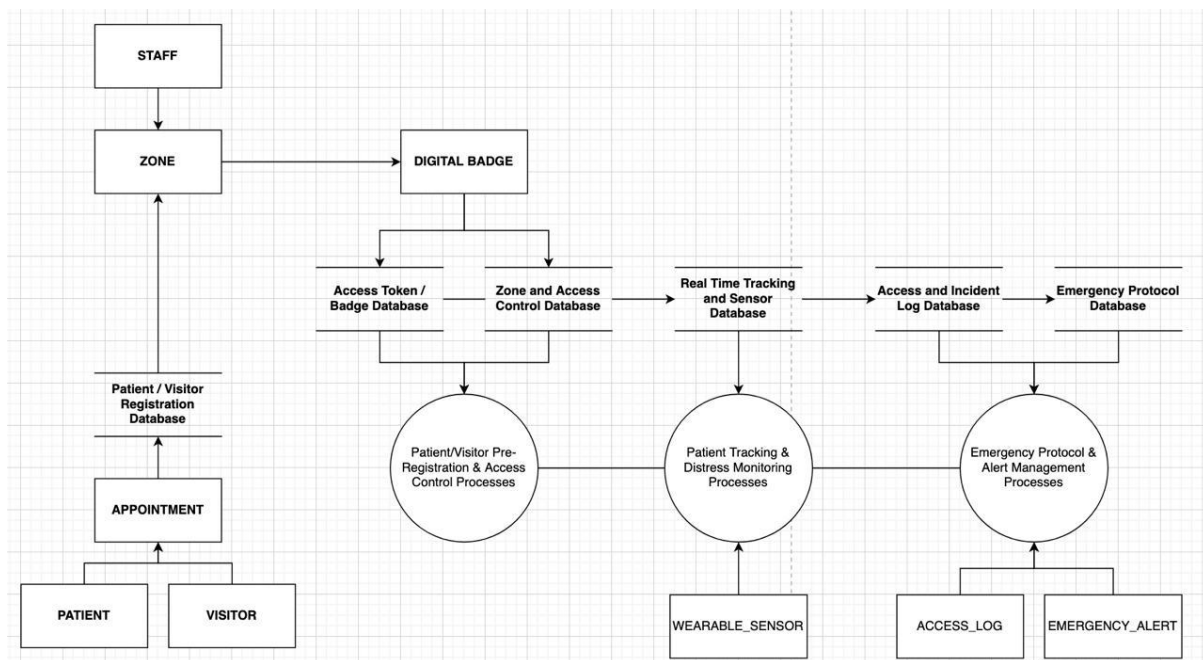


Figure 4 show the Data Flow Diagram

This data flow diagram shows how patients and visitors get register, receive a digital badge, and they receive access to the correct hospital zones. The badge and sensors are then used to track movement and monitor safety in real time. All access events, alerts, and emergencies are recorded in the system so staff can respond quickly. Overall, the diagram explains how information flows through the hospital's new safety and tracking system.

SQL:

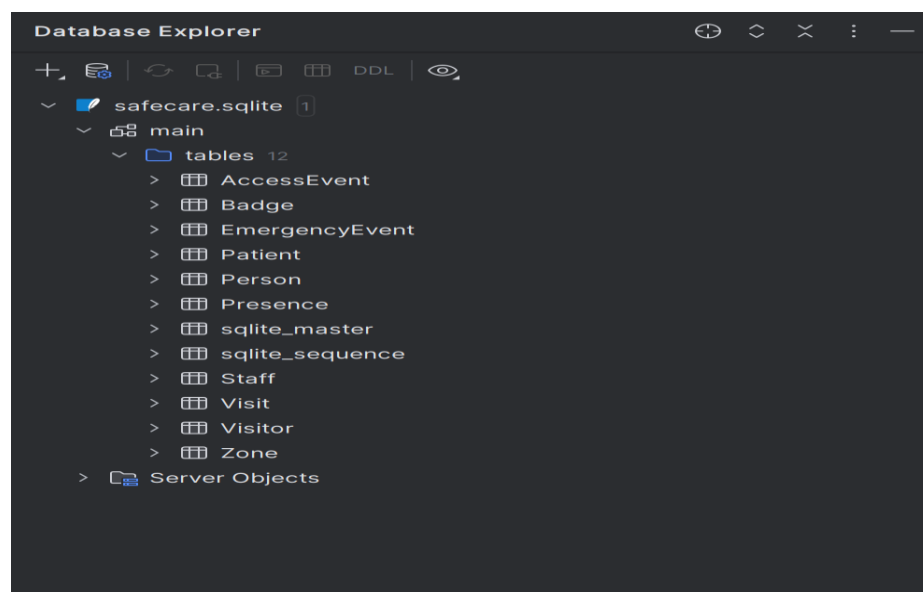


Figure 5 table list from Data Grip

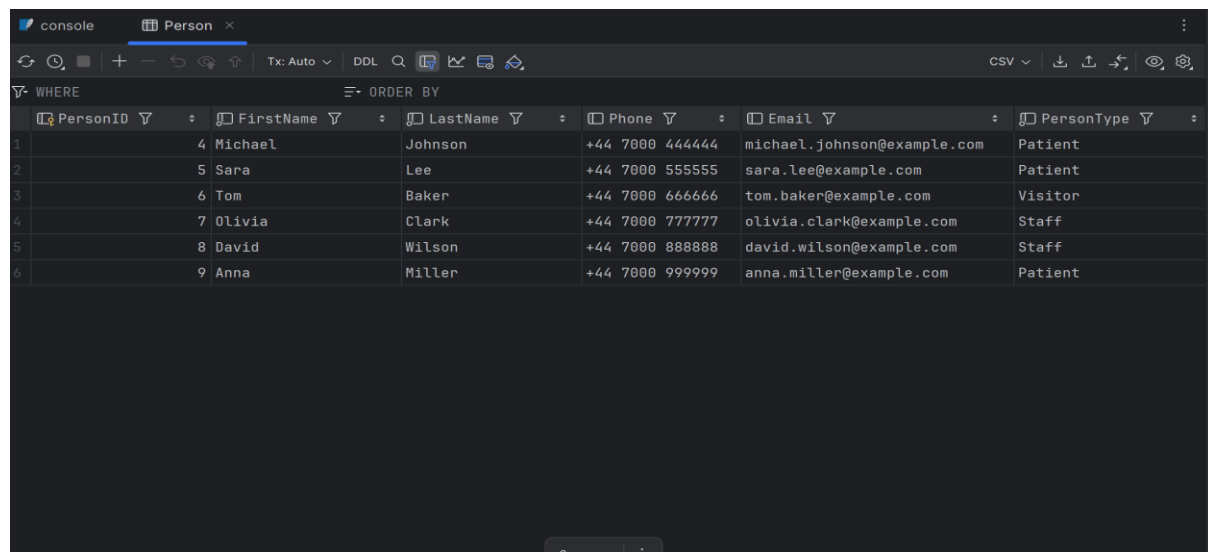
QUERY 1: Add patient/visitor:

```

242 -- QUERY 1: Add patient/visitor
243 -- 1A) Add new Patient via app
244
245 -- Step 1: insert base person record
246 INSERT INTO Person (FirstName, LastName, Phone, Email, PersonType)
247 VALUES ( 'Anna', 'Miller', 'Phone '+44 7000 999999', 'Email 'anna.miller@example.com', 'PersonType 'Patient');
248
249 -- Step 2: link this person as Patient (uses last inserted PersonID)
250 INSERT INTO Patient (PersonID, NHSNumber, RiskFlag)
251 VALUES ( PersonID last_insert_rowid(), 'NHS-3001', 'RiskFlag 0');

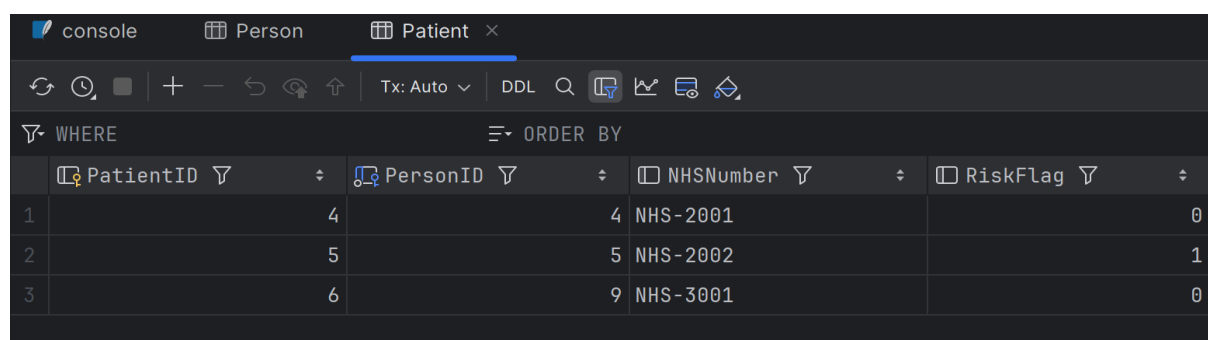
```

Figure 6 SQL query used to add a new patient via the mobile/web app



PersonID	FirstName	LastName	Phone	Email	PersonType
4	Michael	Johnson	+44 7000 444444	michael.johnson@example.com	Patient
5	Sara	Lee	+44 7000 555555	sara.lee@example.com	Patient
6	Tom	Baker	+44 7000 666666	tom.baker@example.com	Visitor
7	Olivia	Clark	+44 7000 777777	olivia.clark@example.com	Staff
8	David	Wilson	+44 7000 888888	david.wilson@example.com	Staff
9	Anna	Miller	+44 7000 999999	anna.miller@example.com	Patient

Figure 7 table “Person” showing the newly added record for Anna Miller



PatientID	PersonID	NHSNumber	RiskFlag
4	4	NHS-2001	0
5	5	NHS-2002	1
6	9	NHS-3001	0

Figure 8 Table “Patient” updated with the new Person ID for Anna Miller

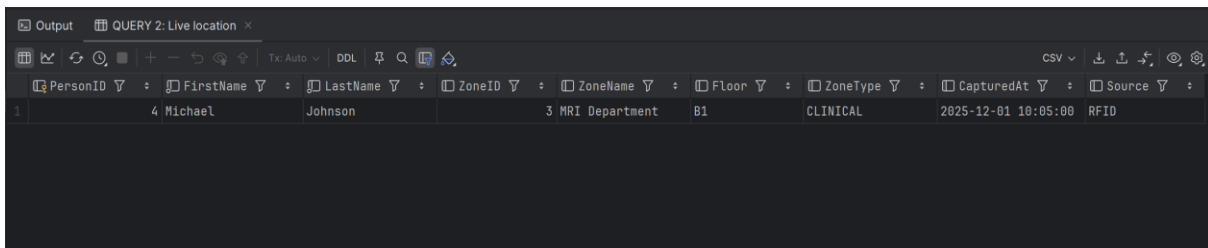
Short explanation:

The system requires that every patient or visitor is first stored in the *Person* table. After the base record is created, the application inserts an additional row in either *Patient* or *Visitor* depending on the user type. The SQL queries in Figure 1 demonstrate this two-step insertion process.

QUERY 2: View live location of a patient or visitor

```
253 -- QUERY 2: Live location
254
255 SELECT
256     per.PersonID,
257     per.FirstName,
258     per.LastName,
259     pr.ZoneID,
260     z.ZoneName,
261     z.Floor,
262     z.ZoneType,
263     pr.CapturedAt,
264     pr.Source
265 FROM Presence pr
266 JOIN Person per 1..n<->1: ON per.PersonID = pr.PersonID
267 JOIN Zone z 1..n<->1: ON z.ZoneID = pr.ZoneID
268 WHERE pr.PersonID = 4
269 ORDER BY pr.CapturedAt DESC
270 LIMIT 1;
```

Figure 9 SQL query used to retrieve the live location



PersonID	FirstName	LastName	ZoneID	ZoneName	Floor	ZoneType	CapturedAt	Source
4	Michael	Johnson	3	MRI Department	B1	CLINICAL	2025-12-01 10:05:00	RFID

Figure 10 result of the query (live location row)

Short explanation:

This query retrieves the patient's or visitor's current (live) location by selecting the most recent presence record based on timestamp. The result shows the zone, floor and the exact capture time of the last detected event.

QUERY 3: View recent emergency events

```

274
275 INSERT INTO EmergencyEvent
276 (PersonID, ZoneID, EmergencyType, Severity, Status, ServiceCalled, TriggeredAt)
277 VALUES
278 ( PersonID 4, ZoneID 3, EmergencyType 'FALL_DETECTED', Severity 'HIGH', Status 'OPEN', ServiceCalled 'MEDICAL_TEAM', TriggeredAt datetime('now'));
279
280 -- 3B View recent emergency events (for reporting / dispatcher screen)
281
282 ✓ SELECT
283 ee.EmergencyID,
284 ee.PersonID,
285 per.FirstName,
286 per.LastName,
287 z.ZoneName,
288 ee.EmergencyType,
289 ee.Severity,
290 ee.ServiceCalled,
291 ee.Status,
292 ee.TriggeredAt
293 FROM EmergencyEvent ee
294 LEFT JOIN Person per 1.n<->0..1 ON per.PersonID = ee.PersonID
295 LEFT JOIN Zone z 1.n<->1..1 ON z.ZoneID = ee.ZoneID
296 ORDER BY ee.EmergencyID DESC
297 LIMIT 5;
298

```

Figure 11 SQL query used to retrieve the 5 most recent emergency events.

Output 3B View recent emerg... / dispatcher screen)

EmergencyID	PersonID	FirstName	LastName	ZoneName	EmergencyType	Severity	ServiceCalled
1	1	<null>	<null>	MRI Department	FIRE	HIGH	FIRE_BRIGADE
2	2	5 Sara	Lee	Pediatrics Ward	FAINTING	MED	MEDICAL_TEAM
3	3	4 Michael	Johnson	MRI Department	FALL_DETECTED	HIGH	MEDICAL_TEAM

Figure 12 Output table showing active emergency events (dispatcher view)

ServiceCalled	Status	TriggeredAt
FIRE_BRIGADE	OPEN	2025-12-01 10:06:00
MEDICAL_TEAM	OPEN	2025-12-01 10:13:00
MEDICAL_TEAM	OPEN	2025-12-01 01:51:24

Figure 13 is a continuation of figure 12

Short explanation:

The system handles emergency calls by inserting a new record into the Emergency Event table.

The INSERT statement is triggered when an incident is detected, automatically recording the patient, zone, severity, service to call and timestamp.

The SELECT statement shows how recent emergency events can be viewed by staff or dispatchers.

Task 5:

UML Diagram:

First Diagram:

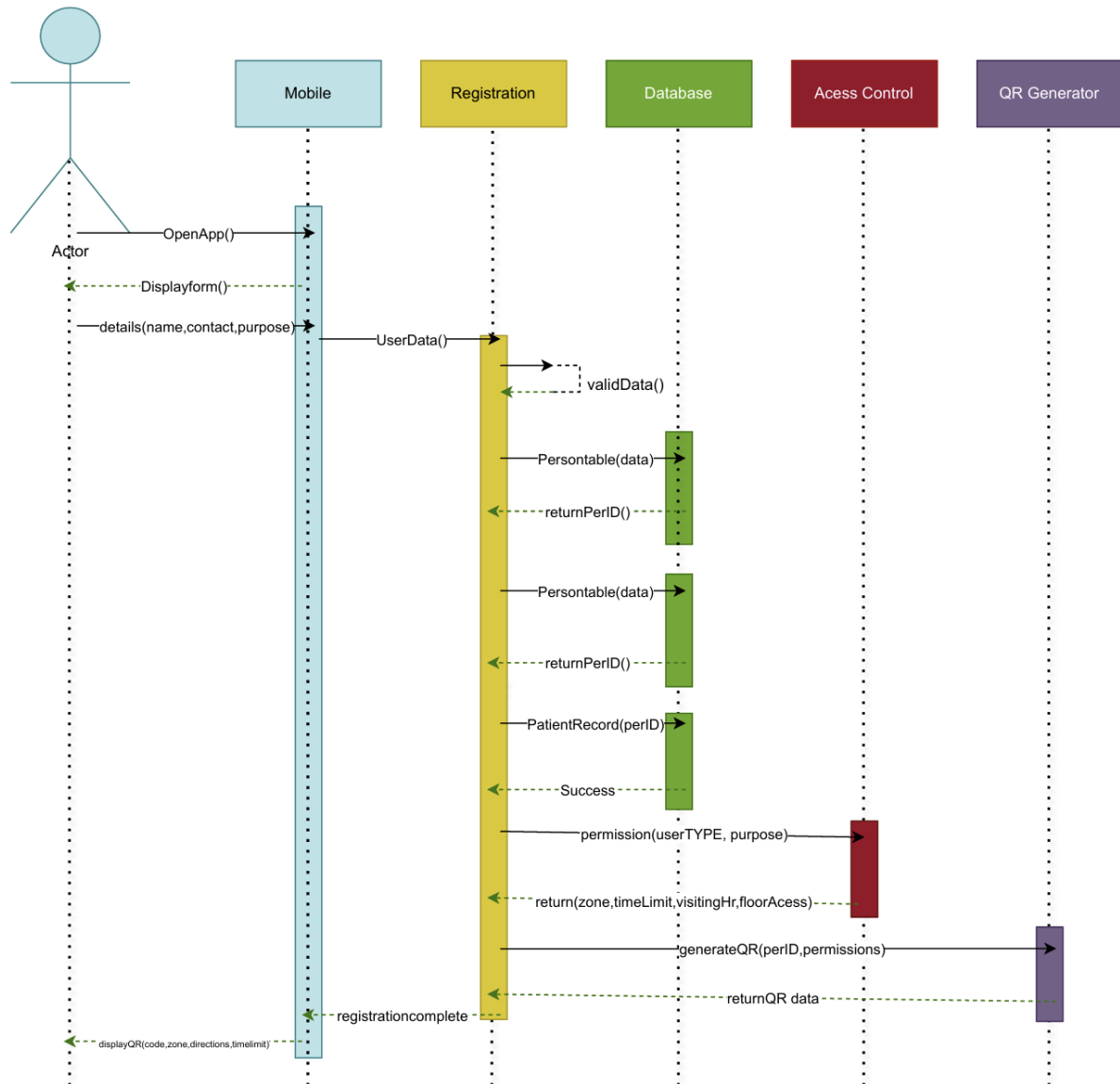


Figure 14 is showing UML Sequence Diagram for Visitor/Patient Registration and QR Code Generation

Second Diagram:

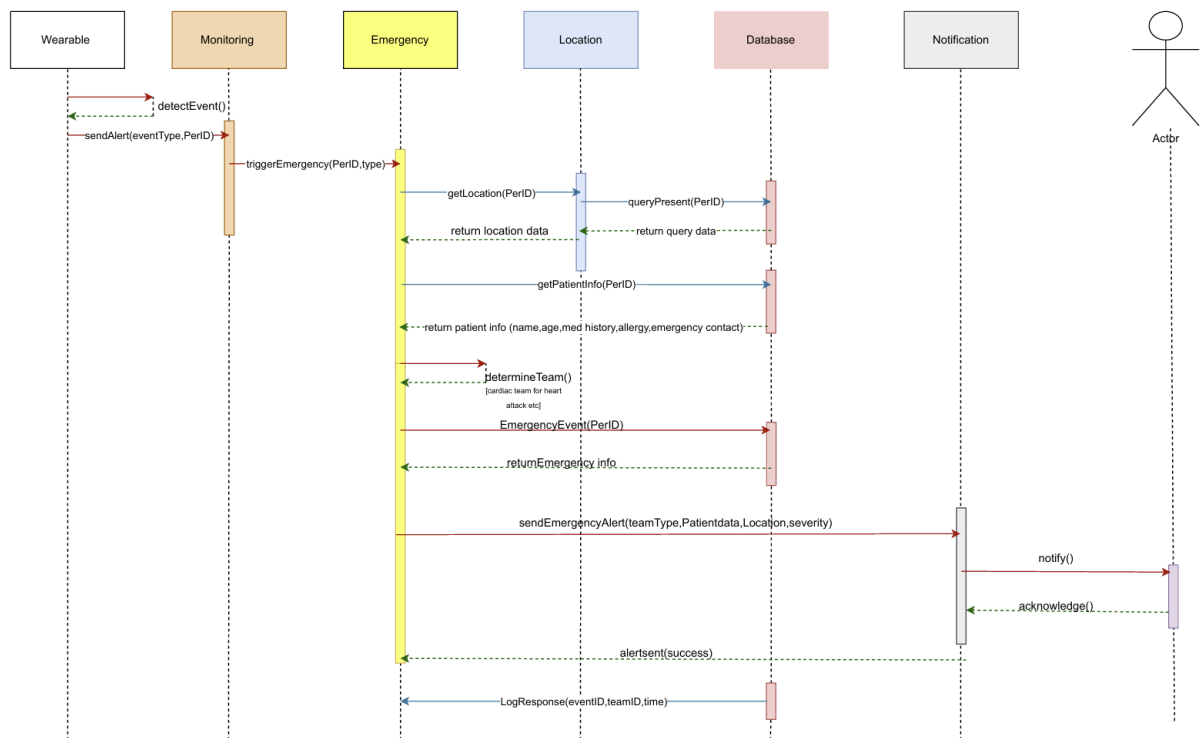


Figure 15 is showing UML Sequence Diagram for Automated Emergency Detection and Response

References list:

- Adenowo, Adetokunbo, and Basirat Adenowo. "Software Engineering Methodologies: A Review of the Waterfall Model and Object- Oriented Approach." International Journal of Scientific & Engineering Research, vol. 4, no. 7, 2013. Accessed 29 Oct. 2025.
- Wiley, John, and Scott Ambler. Agile Modeling: Effective Practices for EXtreme Programming and the Unified Process Wiley Computer Publishing.

Declaration of AI Use

Please append this page to the end of your assignment when you have used AI during the process of undertaking the assignment to acknowledge the ways in which you have used it.

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – YES/NO
- To create an outline of the topic – YES/NO
- To explain concepts – YES/NO
- To support my use of language – YES/NO
- To summarise the following articles/resources: YES/NO